Week 3. By Deng Yufan.

Prob 1. Prove if $a_1, \cdots, a_n$ are in normal form, then so is $a_1 :: a_2 :: \cdots :: a_n :: nil$

Obviously $nil = \lambda x.t = \lambda x.\lambda yz.y$ in normal form.

Induction to prove list l in normal form $\rightarrow$ pair a_i l in normal form.

we have pair a_i l $= \lambda x.x\, a_i\, l$, obviously in normal form.

Prob 2. Show filter is special case of reduce.

We have $filter = \lambda lf.\ reduce\ l\ (\lambda ab.\ ite\ f(a)\ (pair\ a\ b)\ b)\ nil$.

Prob 3. Let $F: [0,1]^n \rightarrow [0,1]$ be a Boolean func. Prove that exist term f that

$f\,[x_1] \cdots [x_n] = [F(x_1, \cdots, x_n)]$. where $[0] = f$. $[1] = t$.

We can write F into Principal Disjunctive Normal Form.

then for ~~every~~ every clause $\bigwedge_{i=1}^{n} x_i / \neg x_i$, we can write a term

~~and~~ and $[(not)\ x_1]\ (and\ [(not)\ x_2]\ (\cdots\cdots (and\ [(not)\ x_{n-1}]\ [(not)\ x_n]))))$ to ~~any~~

represent.

then for $\bigvee_{i=1}^{k} clause_i$, we can write a term

$or\ clause_1\ (or\ clause_2\ (\cdots\ (or\ clause_{k-1}\ clause_k)))$

to represent.

then for the term we just write, we can get

$\lambda x_1 x_2 \cdots x_n.\ term$

to fit the request.

Since and, or, not are simple terms, we finish the proof.

Prob 4. Optional.

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Prob 5. Let $C \subseteq \Sigma^*$ be a lang. Prove ~~是~~ C is Turing-recognizable (RE) iff $\exists$ decidable (R) lang. D st $C = \{x \mid \exists y.\ \langle x,y\rangle \in D\}$.

Insight: let D be a verifier for C. ~~or say~~

If: To solve $C = \{x \mid \exists y.\ \langle x,y\rangle \in D\}$, we can make a NTM.

Step 1. Guess a possible y.

Step 2. Translate input x and guess y into $\langle x,y\rangle$.

Step 3. Simulate TM D with input $\langle x,y\rangle$.

Step 4. If D accept, then accept. Otherwise never accept.

Only if:

To construct a verifier for C, consider

$D = \{\langle x,n\rangle \mid n \in N,\ x$ will be accepted by C in atmost n steps$\}$.

We first prove $C = \{x \mid \exists y.\ \langle x,y\rangle \in D\}$.

If $x \in C$, then the TM C obviously will turn into accept state in finite steps. So $\exists n$ to be a upper bound of steps.

If $\langle x,n\rangle \in D$, then $x \in C$ by def. of D.

Then we prove D is R lang. We make a TM.

Step 1. Translate $\langle x,n\rangle$ to x and n.

Step 2. Simulate TM C with input x.

Step 3. If C ~~halt~~ accept in atmost n steps, then accept.

Step 4. If C take over n steps and not halt, then halt ~~with~~ without accept.

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